

Japan Technology: Hardware – Industrial Electronics: August MOF data (optical fiber/cable) shows sharp rise in ASPs; fundamentals remain

Era of Optics: Innovation behind AI

Insights into optical product technology and trends

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On September 29, the Ministry of Finance released trade statistics for August. Growth in the export value of optical fiber cable was +7% yoy, and that of optical fiber (including some other optical fiber cables) was +140% yoy. **Average selling prices (ASPs) rose sharply, with optical fiber cable up +143% yoy and optical fiber up +112% yoy.**

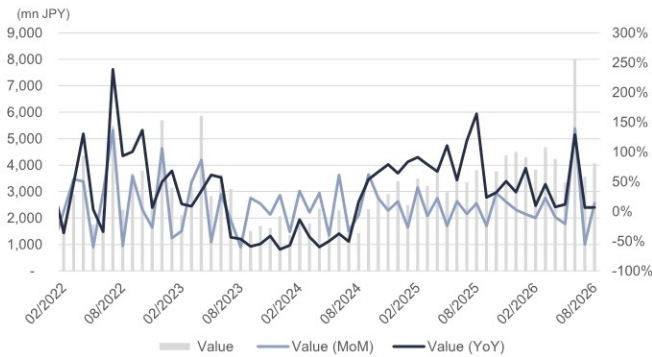
Market participants have expressed concerns about AI investment, including risks related to huge capital outlays on generative AI, power shortages, and the profitability/ROIC of LLMs, as well as opposition to data centers in the key US market. However, as noted in comments at a [Fujikura CEO/CFO call](#) held on September 25, while AI investment is likely to vary and fluctuate in pace by country and region, global AI infrastructure investment looks poised to trend upward. We therefore think it has become important for companies to cast a wide net to capture business opportunities, rather than over-relying on specific countries, regions, or customers. In addition, at Fujikura, we expect a higher weighting for higher-priced MMCs in its high-margin optical connectors. At **Furukawa Electric**, we expect a full-scale launch of thermal (water cooling) products from 2H, as well as price hikes and mix improvement for optical cables, and higher volumes for optical devices (DFB lasers, SOAs, ITLAs). As for **Sumitomo Electric**, while conversations suggest investor interest may be somewhat low at present, it could draw attention for qoq earnings improvement due to a recovery from the prolonged shutdown of a portion of its optical device plant, which was a factor behind the weakness in its infocommunications business in 1Q and its valuation also looks attractive in our view. At **SWCC**, we expect expanded sales of communication cables (e-Ribbon), a larger contribution from semiconductor-related products (probe pins), and we see scope for a resulting multiple re-rating. **Anritsu** stock has been correcting since its 1Q results, partly due to its high multiple, but demand for optical transceiver test equipment looks stronger than expected, and we think 6G may be discussed in the new medium-term plan starting next fiscal year. In fact, major base station manufacturer Ericsson announced it has signed a 6G-related [technology cooperation agreement](#) with Sumitomo Electric and Murata Manufacturing on September 28.

By destination ([Exhibit 7](#)), the data confirmed that the value of exports to the US for

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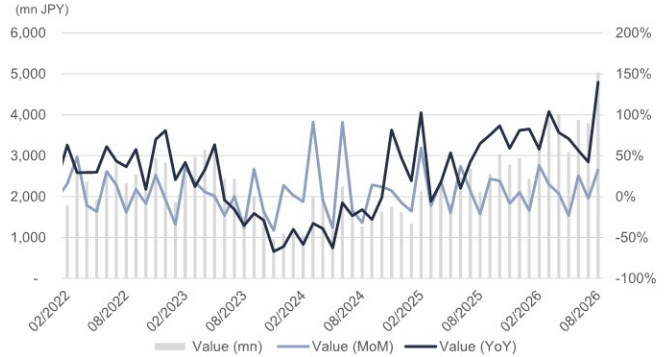
both optical fiber and optical fiber cable increased yoy and mom. By customs jurisdiction ([Exhibit 9](#)), the value of optical fiber cable exports was up yoy at Tokyo and Yokohama. The three major domestic cable manufacturers in our coverage have their main manufacturing bases in Chiba (Fujikura, Tokyo customs), Yokohama (SEI, Yokohama customs), and Mie (Furukawa Electric, Nagoya customs).

Exhibit 1: Export value for optical fiber cable (¥ mn)



Source: Ministry of Finance

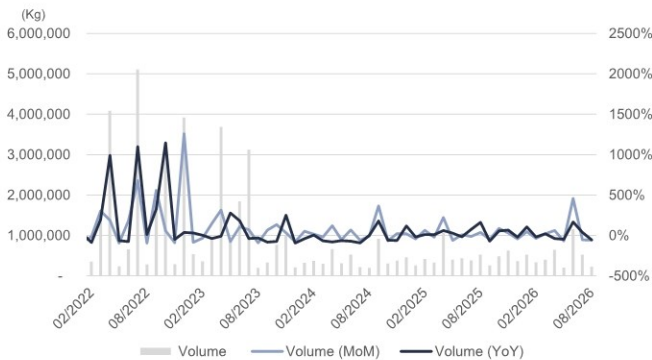
Exhibit 2: Export value for optical fiber (¥ mn)



Includes some other optical fiber cables.

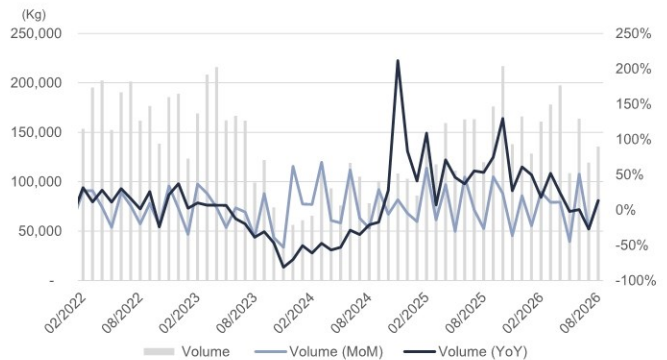
Source: Ministry of Finance

Exhibit 3: Export volume for optical fiber cable (kg)



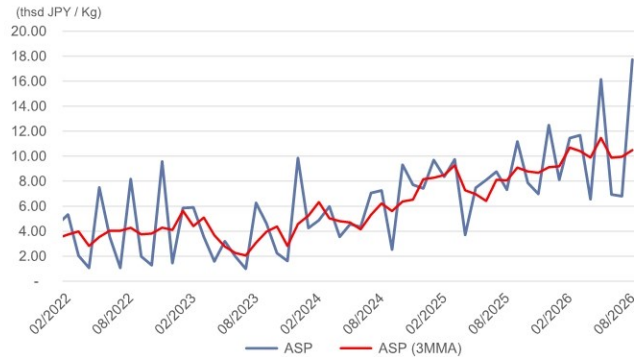
Source: Ministry of Finance

Exhibit 4: Export volume for optical fiber (kg)

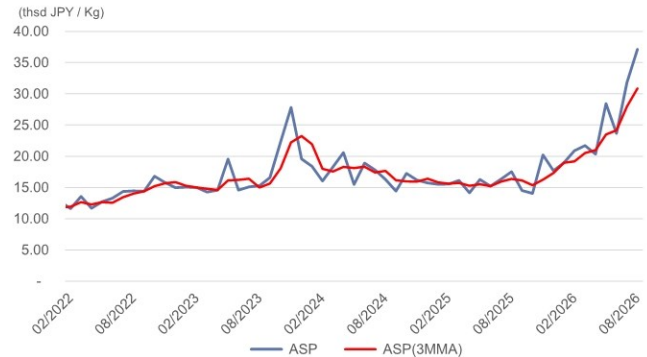


Includes some other optical fiber cables.

Source: Ministry of Finance

Exhibit 5: Average export price for optical fiber cable (¥1,000/kg)

Source: Ministry of Finance

Exhibit 6: Average export price for optical fiber (¥1,000/kg)

Includes some other optical fiber cables.

Source: Ministry of Finance

Exhibit 7: Export value, volume and average export price (yoy, mom) for top 10 destinations (annual export basis)

Region	% in CY2025	Value		Volume		ASP	
		YoY	MoM	YoY	MoM	YoY	MoM
Optical Cable							
US	58%	53%	53%	-7%	50%	65%	2%
Mexico	10%	-99%	-92%	-97%	-78%	-65%	-63%
China	5%	-77%	-83%	-81%	-97%	22%	552%
UK	4%	47%	36%	280%	40%	-61%	-3%
Poland	4%	-	-	-20%	200%	-	-
Taiwan	3%	-82%	-69%	-95%	-96%	301%	736%
Australia	3%	25%	-74%	-66%	-93%	264%	265%
Canada	2%	-10%	175%	-19%	391%	10%	-44%
Philippines	1%	342%	-98%	-68%	-99%	1283%	306%
Vietnam	1%	321%	38%	69%	20%	148%	15%
All regions	100%	7%	14%	-56%	-56%	143%	161%
Optical Fiber							
US	54%	95%	40%	10%	35%	78%	4%
Romania	13%	122%	8%	7%	5%	109%	3%
China	9%	46%	0%	-25%	-35%	93%	53%
Netherlands	3%	-	-	-	-	-	-
Sweden	3%	257%	59%	64%	52%	117%	5%
Turkey	3%	476%	482%	58%	512%	266%	-5%
France	2%	-14%	-33%	42%	-44%	-39%	19%
Taiwan	2%	246%	46%	78%	26%	94%	15%
UK	1%	173%	238%	25%	622%	119%	-53%
UAE	1%	-41%	-	-63%	-	57%	-
All regions	100%	140%	33%	13%	14%	112%	17%

% in CY2025 is based on value for optical fiber cable and volume for optical fiber.

Source: Ministry of Finance

Exhibit 8: Export value, volume and average export price (yoy, mom) for two major European markets and the US

	% in CY2025	Value		Volume		ASP	
		YoY	MoM	YoY	MoM	YoY	MoM
Optical Cable							
US	58%	53%	53%	-7%	50%	65%	2%
UK	4%	47%	36%	280%	40%	-61%	-3%
France	0%	197%	5%	-76%	35%	1154%	-22%
All regions	100%	7%	14%	-56%	-56%	143%	161%
Optical Fiber							
US	54%	95%	40%	10%	35%	78%	4%
UK	1%	173%	238%	25%	622%	119%	-53%
France	2%	-14%	-33%	42%	-44%	-39%	19%
All regions	100%	140%	33%	13%	14%	112%	17%

% in CY2025 is based on value for optical fiber cable and volume for optical fiber.

Source: Ministry of Finance

Exhibit 9: Export value, volume and average export price by customs jurisdiction (yoy, mom)

	% in CY2025	Value		Volume		ASP	
		YoY	MoM	YoY	MoM	YoY	MoM
Optical Cable							
Tokyo	37%	43%	35%	28%	21%	11%	12%
Yokohama	53%	2%	20%	-34%	11%	55%	8%
Nagoya	2%	-63%	134%	-46%	1064%	-31%	-80%
All customs	100%	7%	14%	-56%	-56%	143%	161%
Optical Fiber							
Tokyo	93%	146%	35%	19%	17%	107%	15%
Yokohama	0%	-100%	-100%	0%	-100%	-	-
Nagoya	3%	-55%	6%	-99%	-64%	3750%	198%
All customs	100%	140%	33%	13%	14%	112%	17%

% in CY2025 is based on value for optical fiber cable and volume for optical fiber.

Source: Ministry of Finance

Price Target Risks and Methodology – Fujikura

We are Buy rated. Our 12-month price target of ¥7,600 is based on an EV/EBITDA of 25.0X (FY3/28E; multiple based on the EV/EBITDA and EBITDA margin correlation across its domestic and global competitors). Downside risks: **Telecommunications:** A delay in benefits from hyperscaler investment, a prolonged restraint in telecom carrier investment, and competitors in ultra-high-density optical fiber cables catching up with Fujikura. **Electronics:** A greater-than-expected slowdown in the smartphone market and/or more intense competition. **Automotive products:** Automobile production volume undershooting our assumptions. **Companywide:** Forex swings.

Price Target Risks and Methodology – Furukawa Electric

Our 12-month target price is ¥6,400. We apply an EV/EBITDA multiple of 20X (FY3/28E, based on the correlation between EV/EBITDA and EBITDA growth rate for domestic and overseas competitors). We are Buy rated. Risks: **Communications solutions:** If sales expansion of high-margin optical products for data centers does not proceed as expected and there is no improvement in profits. If the profitability of the acquired Furukawa FITEL Optical Components (FFOC) deteriorates again due to factors such as changes in competitiveness. If generative AI data center investment slows for reasons such as delays in data center construction due to supply shortages of some essential

products, or if investment in generative AI data centers starts to wane as project profitability deteriorates due to price hikes for component products. **Energy infrastructure:** If demand for both high-voltage and medium/low-voltage cables slows due to factors such as a slowdown in generative AI investment. If the company decides on capex for HVDC cables but fails to secure the expected orders from customers. **Electronics & automotive systems:** If auto production volume or sales of models supplied by the company do not grow as much as expected. If fluctuations in auto production volume are larger than expected and the company is unable to absorb fixed costs through measures such as price revisions. If material prices surge more than expected and the company is unable to pass these on to customers. **Functional products:** If, in the copper foil business, sales expansion of high-function circuit foils does not proceed as planned and profitability improvement is not expected, if orders for thermal products for generative AI are not booked as sales for some reason or are delayed, or if the acquisition of new customers for semiconductor manufacturing tape does not progress as expected.

Price Target Risks and Methodology – Sumitomo Electric Industries

We are Buy rated. Our 12-month target price is ¥3,300. We apply a target EV/EBITDA multiple of 13.0X (based on FY3/28E, using an SOTP approach). Key downside risks:

Automotive segment: Decline in automobile production volume; slower-than-expected pass-through of material procurement costs to selling prices. **Infocommunications segment:** Longer-than-expected inventory adjustments due to a slowdown in hyperscaler capex; catch-up by competitors in ultra-high density optical fiber cables. **Environment & energy segment:** Delays in new and replacement power infrastructure projects due to difficulties in procuring materials and securing workers; slower-than-expected earnings contributions from the Japanese government's clean energy strategy. **Industrial materials and other segments:** Slowdown in demand for cemented carbide tools, etc., due to a macroeconomic slowdown; increase in manufacturing costs for sintered parts, etc., due to rising energy prices. **Company-wide:** FX fluctuations; sustained high material prices.

Price Target Risks and Methodology – SWCC

Valuation methodology: We are Buy rated. Our 12-month target price of ¥3,340 is based on an EV/EBITDA of 11.5X (FY3/28E, multiple based on the correlation between EV/EBITDA and EBITDA margin for global competitors).

Key downside risks: **Energy & infrastructure segment:** The balancing out of supply and demand for wires/cables and power equipment in Japan, resulting in lower selling prices; a lack of progress with the further leveling of construction work and improvement of personnel allocation efficiency, resulting in slower-than-expected profit margin improvement; an influx of foreign workers significantly alleviating the shortage of construction workers in Japan. **Communication & components segment:** Sharp declines in optical fiber cable selling prices due to a large supply of inexpensive products from China, India, and other countries, and profitability deteriorating as a result; home appliance sales in China and Japan falling short of expectations; acceleration in the trend toward a paperless society, resulting in a steeper-than-expected decline in sales of printer rollers. A slower-than-expected shift to EVs by the company's customers, leading to lower demand; significant declines in automobile production volume, including EVs; increases in prices of materials such as copper, with the company unable

to pass these increases on to customers. **Company-wide:** The implementation of large-scale growth investments aimed at furthering overseas expansion in the electronic equipment & components and communication & industrial devices segments, with returns on these investments falling short of expectations and significantly impacting company-wide margins; yen appreciation.

Price Target Risks and Methodology – Anritsu

Our 12-month target price is ¥5,400, based on a target EV/EBITDA multiple of 17X on FY3/28E (our target multiple is based on the correlation between EV/EBITDA and EBITDA margin for global competitors). We are Buy rated.

Risks: **Test and Measurement:** Demand for 5G applications has already cooled, and there is a risk that sales may not grow as expected, leading to a deterioration in profit margins. For the data center business, uncertainty due to US tariffs could lead to slower decision-making by customers, resulting in stagnating or weaker demand. Testing methods may change from 100% testing to sampling testing. **PQA:** Possible decline in domestic market share, or the failure of sales expansion to proceed as expected in regions where the company is accelerating business development, such as North America, Europe, and China. **Environmental Measurement:** The possibility that earnings in battery test equipment fail to grow as expected due to a slowdown in the BEV market. **Company-wide:** Disappointing returns from M&A and/or growth investment. An increase in the R&D expense ratio. Yen appreciation.

Disclosure Appendix

Reg AC

We, Ryo Harada and Hiroki Muramatsu, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Ryo Harada Goldman Sachs Japan Co., Ltd., Hiroki Muramatsu Goldman Sachs Japan Co., Ltd..

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

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Goldman Sachs had a non-securities services client relationship during the past 12 months with: Anritsu (¥3,377), Fujikura (¥4,931), Furukawa Electric (¥3,684) and Sumitomo Electric Industries (¥2,174)

Distribution of ratings/investment banking relationships

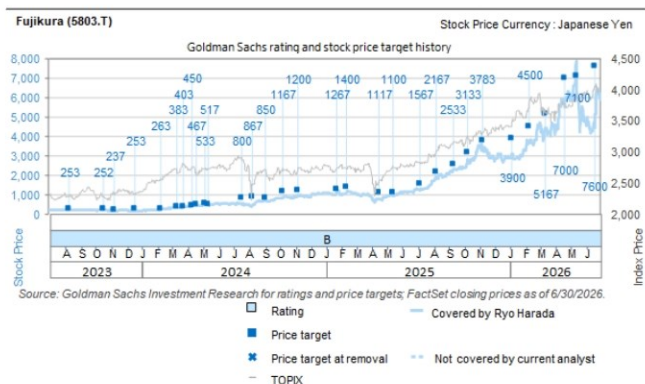
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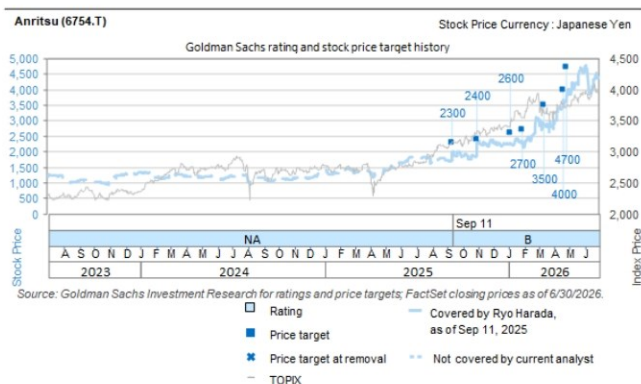
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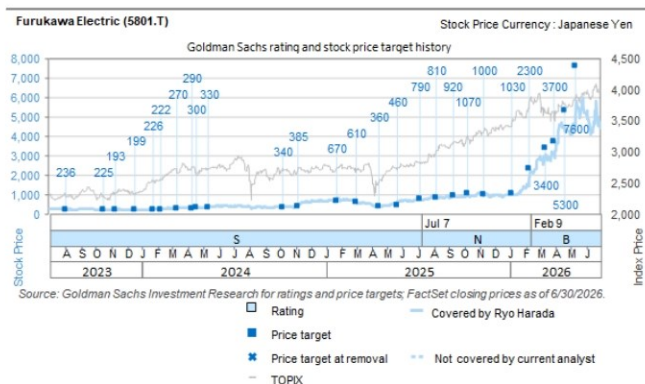
Price target and rating history chart(s)



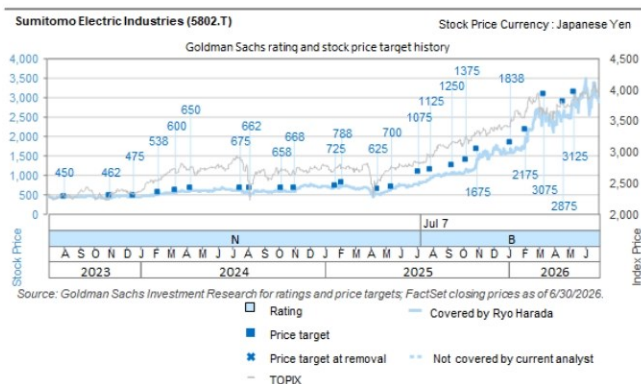
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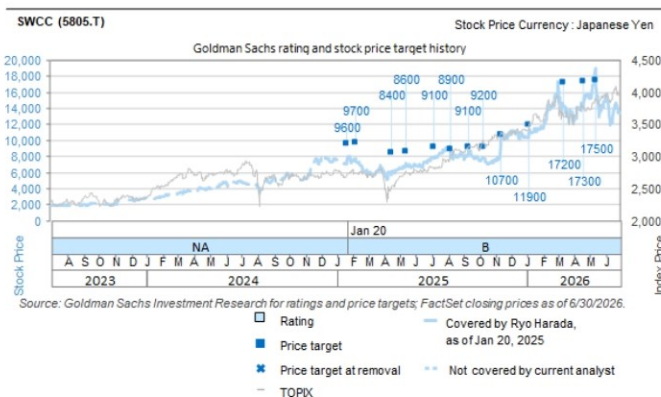
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Target price history table(s)

Fujikura (5803.T)

Date of report	Target price (¥)	Closing price (¥)
07-Aug-26	7,600	5,185

Sumitomo Electric Industries (5802.T)

Date of report	Target price (¥)	Closing price (¥)
31-Jul-26	3,300	2,184

Fujikura (5803.T)

Date of report	Target price (¥)	Closing price (¥)
07-Jul-26	7,500	5,072
18-Jun-26	7,600	4,461
14-May-26	7,100	6,355
20-Apr-26	7,000	5,529
13-Mar-26	31,000	4,420
09-Feb-26	27,000	3,659
05-Jan-26	23,400	3,074
07-Nov-25	22,700	3,403
09-Oct-25	18,800	2,798
10-Sep-25	15,200	2,272
07-Aug-25	13,000	1,918
07-Jul-25	9,400	1,235
13-May-25	6,600	957
16-Apr-25	6,700	761
10-Feb-25	8,400	1,113
22-Jan-25	7,600	1,157
07-Nov-24	7,200	968
07-Oct-24	7,000	831
02-Sep-24	5,100	736
08-Aug-24	5,200	423
18-Jul-24	4,800	518
13-May-24	3,100	474
06-May-24	3,200	463
18-Apr-24	2,800	444
09-Apr-24	2,700	422
24-Mar-24	2,420	358
12-Mar-24	2,300	316
08-Feb-24	1,580	210
19-Dec-23	1,520	174
08-Nov-23	1,420	192
17-Oct-23	1,510	200

Sumitomo Electric Industries (5802.T)

Date of report	Target price (¥)	Closing price (¥)
07-Jul-26	3,400	2,443
12-May-26	12,500	2,913
20-Apr-26	11,500	2,494
13-Mar-26	12,300	2,598
03-Feb-26	8,700	1,914
05-Jan-26	7,350	1,678
31-Oct-25	6,700	1,413
09-Oct-25	5,500	1,158
10-Sep-25	5,000	1,073
31-Jul-25	4,500	940
07-Jul-25	4,300	756
13-May-25	2,800	658
16-Apr-25	2,500	516
04-Feb-25	3,150	752
22-Jan-25	2,900	727
01-Nov-24	2,670	583
07-Oct-24	2,630	606
07-Aug-24	2,650	530
18-Jul-24	2,700	600
09-Apr-24	2,600	602
12-Mar-24	2,400	556
05-Feb-24	2,150	501
19-Dec-23	1,900	443
02-Nov-23	1,850	404

Furukawa Electric (5801.T)

Date of report	Target price (¥)	Closing price (¥)
06-Aug-26	6,400	3,909
07-Jul-26	7,200	3,674
12-May-26	76,000	5,043
20-Apr-26	53,000	4,307
30-Mar-26	37,000	3,095
13-Mar-26	34,000	3,069
09-Feb-26	23,000	1,750
05-Jan-26	10,300	1,054
10-Nov-25	10,000	946
09-Oct-25	10,700	986
10-Sep-25	9,200	925
07-Aug-25	8,100	907
07-Jul-25	7,900	697
22-May-25	4,600	624
16-Apr-25	3,600	418
03-Mar-25	6,100	623
22-Jan-25	6,700	792
07-Nov-24	3,850	403
07-Oct-24	3,400	362
13-May-24	3,300	345
18-Apr-24	3,000	345
09-Apr-24	2,900	333
12-Mar-24	2,700	292
08-Feb-24	2,220	277

SWCC (5805.T)

Date of report	Target price (¥)	Closing price (¥)
07-Aug-26	16,700	12,960
07-Jul-26	16,200	11,910
14-May-26	17,500	16,990
20-Apr-26	17,300	14,090
13-Mar-26	17,200	14,230
05-Jan-26	11,900	10,960
12-Nov-25	10,700	9,450
09-Oct-25	9,200	7,880
10-Sep-25	9,100	8,300
08-Aug-25	8,900	7,920
07-Jul-25	9,100	7,800
13-May-25	8,600	7,160
16-Apr-25	8,400	5,850
07-Feb-25	9,700	7,310
20-Jan-25	9,600	7,080

Furukawa Electric (5801.T)

Date of report	Target price (¥)	Closing price (¥)
24-Jan-24	2,260	253
19-Dec-23	1,990	217
09-Nov-23	1,930	232
17-Oct-23	2,250	228

SWCC (5805.T)

Date of report	Target price (¥)	Closing price (¥)
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Anritsu (6754.T)

Date of report	Target price (¥)	Closing price (¥)
30-Jul-26	5,400	3,553
07-Jul-26	5,300	4,056
27-Apr-26	4,700	3,755
20-Apr-26	4,000	3,466
13-Mar-26	3,500	2,891
29-Jan-26	2,700	2,369
05-Jan-26	2,600	2,288
30-Oct-25	2,400	1,887
11-Sep-25	2,300	1,745

Price targets shown in table(s) are unadjusted for corporate actions.

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